

SET A –Solutions

1. Use Google's **site operator** (e.g., `site:example.com`) to view indexed pages.
The result count shows how many pages are indexed.
2. CSRF token is a **random secret value** included in each request.
The server verifies it, ensuring the request is intentional and not forged.
3. Largest costs: **Marketing (≈35%) and Content Development (≈30%)**.
These dominate over hardware/software costs.
4. Ranking signals include: **relevance, content quality, and backlinks (authority)**.
Page speed and mobile usability also matter.
5. GA4 uses **JavaScript tracking code in the user's browser**.
It collects user interactions and sends data to Google servers.
6. CSRF: **forces user actions using authentication cookies**.
XSS: **injects malicious scripts into the browser**.
7. Examples: **slow loading, pop-ups, confusing navigation, forced registration**.
These create friction and reduce user satisfaction.
8. Three factors: **page content, page generation, page delivery**.
They ensure fast and efficient website performance.
9. Vulnerability Assessment: **identifies weaknesses only**.
Penetration Testing: **actively exploits them to assess impact**.

SET B – Solutions

1. Website optimization depends on **content, generation, and delivery**.
Each affects speed and user experience.
2. Key signals: **relevance, authority (links), and content quality**.
3. CSRF token ensures **request includes valid session-matching token**.
Without it, server rejects the request.
4. Use **site:domain.com** in Google search.
It shows indexed pages count.
5. Issues: **pop-ups, slow speed, broken links, poor navigation**.
6. Vulnerability Assessment: **finds and prioritizes risks**.
Pen Testing: **simulates real attacks by exploiting them**.
7. Major costs: **marketing and content development**.
8. GA4 collects data via **client-side JavaScript tracking**.
Data is sent to Google servers.
9. XSS: **runs malicious scripts in browser**.
CSRF: **tricks browser into sending unwanted requests**.

SET C – Solutions

1. GA4 uses **tracking code in browser to capture clicks, views, and interactions**.
Data is transmitted to analytics servers.

2. Use **Google site operator (site:domain)**.
Displays indexed page count.

3. CSRF token works because it is **secret, random, and verified by server**.
Attackers cannot guess or access it.

4. Ranking signals: **relevance, authority (backlinks), performance**.

5. Budget dominated by **marketing and content development**.

6. CSRF targets **server by abusing user session**.
XSS targets **user by executing malicious scripts**.

7. Optimization factors: **content, generation, delivery**.

8. Vulnerability Assessment: **identification only**.
Pen Testing: **exploitation and impact analysis**.

9. Examples: **slow loading, confusing UI, pop-ups, lack of search feature**.

SET D – Solutions

1. Vulnerability Assessment **finds and prioritizes vulnerabilities**.
Penetration Testing **exploits them to evaluate real risk**.
2. Ranking factors: **relevance, content quality, authority (links)**.
3. GA4 uses **client-side JavaScript to collect user data**.
Data is sent to Google, not processed on the web server.
4. Use **Google search: `site:domain.com`**.
5. CSRF token ensures **request contains valid token tied to session**.
Invalid requests are rejected.
6. Examples: **slow pages, pop-ups, confusing navigation, forced login**.
7. Largest costs: **marketing and content development**.
8. Optimization requires **content, generation, and delivery improvements**.
9. XSS: **injects scripts and executes in browser**.
CSRF: **forces authenticated user actions without consent**.